

What to Expect When Using DECT NR+

Fabian Graf, Thomas Watteyne, and Michael Villnow

ABSTRACT

DECT NR+ is the new kid on the block in wireless technologies: it re-purposes the 1.9 GHz Digital Enhanced Cordless Telecommunications (DECT) standard for IoT-type applications. How does DECT NR+ perform in practice and where should we be using it? The ambition of this article is to provide answers to those questions. We start by an overview of the fundamental principles of the physical layer. We then survey the DECT NR+ products on the market today. Using the nRF91 series from Nordic Semiconductor, we conduct a comprehensive set of hands-on power consumption and communication range measurements. Our results place DECT NR+ in a gap in-between existing technologies. Its range is comparable to long range standards such as IEEE 802.15.4g under certain parameter choices: 200 m in an urban setting, 6 km in the most favorable line-of-sight conditions. For higher order Modulation and Coding Schemes, range drops and is rather comparable to Wi-Fi. The nRF9161 draws significant power (at 3.7 V, 220 mA transmitting at +19 dBm, 45 mA receiving), approximately 10× higher than BLE radios. This is likely primarily due to early stage design inefficiencies and the inherent complexity of the DECT NR+ physical layer, limiting its adoption in battery-powered applications. We conclude that DECT NR+ is particularly appropriate for applications that require a dynamic tradeoff between communication range and data rate, but are not cost-sensitive.

INTRODUCTION

Wireless connectivity is at the heart of Internet of Things (IoT) applications, allowing devices to communicate without the constraints of a wired infrastructure. The scope of 5G standardization has expanded into IoT applications [1]. While cellular 5G networks are well-suited for high-bandwidth and lowlatency use cases, their infrastructure-heavy nature and operational costs make them less ideal for widespread IoT deployment, especially in industrial and private networks. To bridge the gap between traditional cellular and unlicensed wireless IoT solutions, DECT-2020 NR, also known as “DECT NR+”, is announced as “the first non-cellular 5G standard” officially recognized by the International Telecommunication Union Radiocommunication Sector (ITU-R). The promise is that DECT NR+ avoids costly contracts and dependencies associated with licensed bands, unlike cellular technologies such as 5G NR, LTE-M or Narrowband IoT (NB-IoT). DECT NR+ is specified by European Telecommunications Standards Institute (ETSI)

and designed to support the ultra-reliable lowlatency communication (URLLC) [2] and massive machine-type communication (mMTC) pillars of 5G [3]. One of its key features is its operation in the 1.9 GHz license-free frequency band, traditionally reserved for Digital Enhanced Cordless Telecommunications (DECT) phones, offering the advantage of minimal external interference. DECT NR+ potentially enables self-organizing mesh networks, long-range communication, and a wide range of adjustable parameters, offering lots of flexibility. As a newly emerging standard, DECT NR+ has sparked significant interest. With promises of scalable connectivity, DECT NR+ is being considered for smart city deployments, environmental and agricultural monitoring, and even high-throughput scenarios like audio streaming [4]. This raises the question: *Does DECT NR+ really deliver in practice?*

This article serves as a primer on DECT NR+, aimed at researchers and advanced engineers seeking hands-on insights and an intuition about what to realistically expect from the technology. We present a first real-world evaluation of DECT NR+ in terms of power consumption and range, complementing related work that has focused on theoretical models and simulations [5, 6]. We carry out a comprehensive series of range measurements using commercial off the shelf (COTS) hardware from Nordic Semiconductor. In indoor environments such as offices and factory halls, we observe good performance, where even low transmit powers below 0 dBm are sufficient to maintain a high Packet Delivery Ratio (PDR). In outdoor line-of-sight (LOS) conditions, we are able to push the system to its limits and achieve communication over distances of up to 6 km. However, in urban non-LOS scenarios, the reliable communication range drops significantly down to around 200 m, highlighting the challenges posed by dense, obstructed environments. We also present detailed energy consumption measurements, offering insights into the operational costs of DECT NR+ and identifying parameter settings under which it can still be considered a low-power technology. For example, transmitting a 249 byte packet using Modulation and Coding Scheme (MCS) 0 at +19 dBm results in a substantial charge drawn of approximately 700 μ C. Conversely, when choosing a higher-order MCS, the energy consumption per bit drops. We emphasize that the classification as a low-power technology is heavily dependent on the design of the upper-layer protocols and the overall system architecture. Nevertheless, since the Physical (PHY)



layer represents the fundamental physical limitation that cannot be overcome by nature, and efforts in developing upper-layer protocols are still in their early stages, we focus exclusively on PHY experiments in this article. By delivering the first benchmarking of DECT NR+ performance in practical scenarios, this article serves as a decision-making guide for IoT users who are reaching data rate or range limitations and are considering DECT NR+ as an alternative.

The article is organized as follows. A primer on the DECT NR+ PHY layer, details achievable payload sizes and data rates. We provide an overview of the current hardware platforms and software stacks available for DECT NR+. We present the results of power consumption measurements, offering insights into the rationale behind parameter selection and its impact on energy efficiency. We explore various range test environments, and evaluates the performance of DECT NR+ in terms of PDR, through a comprehensive set of real-world representative measurements. We discuss what to expect when using DECT NR+ and how it positions itself among other wireless communication standards.

THE DECT NR+ PHYSICAL LAYER: A PRIMER

DECT-2020 NR is specified in the ETSI TS 103 636 standard [7]. We refer to release 2 from October 2024, which comprises the specification of radio reception and transmission requirements (Part 2), the Physical layer (Part 3), the Medium Access Control (MAC) layer (Part 4) and the Data Link Control (DLC) and Convergence layers (Part 5).

MODULATION

DECT NR+ uses Cyclic Prefix (CP) Orthogonal Frequency Division Multiplexing (OFDM). OFDM is a modulation technique that transmits data in parallel over multiple closely spaced subcarriers to make it more robust against frequencyselective fading. The CP is a copy of the end of an OFDM symbol appended to its beginning, serving as a guard interval to mitigate inter-symbol interference and preserve subcarrier orthogonality. There are three major configuration parameters in DECT NR+ regarding modulation with OFDM.

The subcarrier scaling factor $\mu \in \{1, 2, 4, 8\}$ determines the spacing between two adjacent orthogonal subcarriers. As μ increases, the OFDM symbol duration becomes shorter, leading to a higher required bandwidth. Therefore, latency decreases and faster transmission is possible, allowing for higher robustness against Doppler shifts. However, the shorter CP reduces protection against multipath effects.

The Fourier transform scaling factor $\beta \in \{1, 2, 4, 8, 12, 16\}$ determines the number of subcarriers. Larger values of β lead to more subcarriers and increase the available channel bandwidth and, therefore, the data rate capacity at the cost of larger Fast Fourier Transform (FFT) sizes. Although this means that the PHY can achieve higher data rates, processing power rises as the transceiver complexity grows [8].

The MCS defines the modulation used for the subcarrier symbols and the code rate of the turbo code used for Forward Error Correction (FEC). The modulation scheme indicates how many bits per symbol can be transmitted on each subcarrier. The

code rate R characterizes the fraction of bits in the Physical Data Channel (PDC) that are actually used as information bits, whereas the rest is used as redundancy to recover from transmission errors. The standard specifies in total 12 different MCS, with the first five utilizing Binary Phase-Shift Keying (BPSK), Quadrature Phase-Shift Keying (QPSK) and 16-Quadrature Amplitude Modulation (QAM):

- MCS 0: BPSK, $R = 0.50$,
- MCS 1: QPSK, $R = 0.50$,
- MCS 2: QPSK, $R = 0.75$,
- MCS 3: 16-QAM, $R = 0.50$,
- MCS 4: 16-QAM, $R = 0.75$.

As the MCS order increases, larger payload sizes fit into a packet, but robustness decreases.

REGULATIONS, FREQUENCY AND TRANSMIT POWER

When DECT was standardized for cordless phones in 1992, frequency bands were set aside for this application. These are unlicensed bands: neither the vendor nor the end user need to pay a fee to use them. Since DECT NR+ is under the DECT umbrella, it inherits from these license-free attributes, making it attractive for technology providers in the same way that Wi-Fi, Bluetooth and IEEE 802.15.4 operate in license-free bands. And because the DECT NR+ frequency bands are less crowded than the 2.4 GHz bands (where Wi-Fi and Bluetooth operate), the promise is that DECT NR+ deployments suffer less from external interference, at least early in the adoption. The frequency band designated for DECT is not globally standardized, and in some countries, such as China, no designated band is available.

The standard defines 22 frequency bands. European national regulations mandate the use of Band 1 (1880–1900 MHz), which contains 21 channels (numbered 1657 to 1677), with center frequencies spaced by 0.864 MHz. In North America, Band 9 (1920–1930 MHz) must be used. It contains 9 channels (numbered 1703 to 1711), also spaced by 0.864 MHz.

The DECT NR+ standard specifies that transmission power levels for Class I devices — which typically serve as gateways, base stations, or outdoor nodes for long-range communication — must not exceed +23 dBm. This aligns with local regulations for the DECT frequency bands, allowing a peak effective radiated power (ERP) of 250 mW (+23.98 dBm).

Unlike sub-GHz bands, we are unaware of regulations that limit duty-cycle or impose a minimal inter-frame spacing.

PACKET STRUCTURE AND DATA RATES

Slots are the unit used for defining packet lengths of a radio frame in DECT NR+, rather than bytes. The standard mandates that the duration of a frame cannot exceed 10 ms, corresponding to a maximum of $N_{\text{slot}}^{\text{FRAME}} = 24$ slots.

Each slot is further divided into subslots, and for a subcarrier spacing factor of $\mu = 1$, one slot consists of $N_{\text{subslot}}^{\text{SLOT}} = 2$ subslots. A subslot always consists of $N_{\text{symbol}}^{\text{SUBSLOT}} = 5$ OFDM symbols. Each OFDM-symbol is split up on a number of subcarriers, depending on the parameter β . For $\beta = 1$, the PHY uses 64 subcarriers. This results in a matrix structure: columns represent OFDM symbols, rows subcarriers. The individual cells of this matrix, known as resource elements (REs), form the small-

The CP is a copy of the end of an OFDM symbol appended to its beginning, serving as a guard interval to mitigate inter-symbol interference and preserve subcarrier orthogonality.

MCS/Subslots	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
MCS 0	0	17	33	50	67	83	99	115	133	149	165	181	197	213	233	249
MCS 1	4	37	69	103	137	169	201	233	263	295	327	359	391	423	463	495
MCS 2	7	57	107	157	205	253	295	343	399	447	495	540	596	644	692	740
MCS 3	11	77	141	209	271	335	399	463	532	596	660	732	793	857	929	993
MCS 4	18	117	217	311	407	503	604	700	801	897	1001	1094	1198	1291	1395	1491

TABLE 1. Payload size in bytes of the PDC as a function of MCS and the number of subslots computed for $\mu = 1$ and $\beta = 1$. Configurations leading to sizes larger than 700 bytes are not supported by the nRF91 and therefore marked in *italics*.

est unit of the radio frame. Within these REs, the information bits, referred to as the PDC, are placed.

Not all REs are used for the PDC. This is because, depending on the parameter selection and number of slots, REs are also used for the Synchronization Training Field (STF), Demodulation Reference Signal (DRS), Physical Control Channel (PCC) and Guard Band (GB). The conversion from packet duration in slots/subslots into bytes is therefore not straightforward, as it follows a set of parameter-dependent formulas. Table 1 is a quick reference. It contains a number of conversions for $\mu = 1$ and $\beta = 1$, for a packet length of up to 8 slots, and for MCS 0 to MCS 4.

The computation of the data rates from the values in Table 1 is straightforward. For $\mu = 1$, the duration of a single OFDM symbol is 41.667 μ s. As an example, a packet length of 8 subslots and MCS 4 yields an achievable transport block size of 700 B. Since $N_{\text{subslot}}^{\text{symbol}} = 5$ OFDM symbols fit into 1 subslot, the transmission of those 700 B (5600 bits) in 8 subslots takes 1.666 ms, resulting in a theoretical data rate of 3.36 Mbit/s.

WHAT IS AVAILABLE ON THE MARKET TODAY?

Today, only two vendors offer DECT NR+-compliant chipsets. The nRF91 series from Nordic Semiconductor features three chips: nRF9161, nRF9151, nRF9131. Last Mile Semiconductor commercializes the LM10XX system-on-chip.

To explore the PHY, we use the Nordic Semiconductor nRF91 series. Their Connect SDK software package includes basic sample applications based on Zephyr, which we use as a base for our experiments.

The nRF9161 was designed for cellular IoT, and supports LTE-M and NB-IoT. Its radio front-end is flexible enough that, by updating the core firmware on its radio co-processor, the chip switches over to DECT NR+. The nRF9151 is nRF9161's successor, with higher output power. The nRF9131 is a cheaper version, at the cost of higher power consumption.

For all of the mentioned chips, dedicated Development Kits (DK) or Evaluation Kits (EK) are available to support quick prototyping and experimentation. Both the nRF9161-DK and nRF9151-DK feature a PCB antenna. While they are functional, these boards are not designed for maximum range so are not optimal for range testing. Instead, we used the nRF9131-EK that features an external antenna connector. The range measurement results we get with the nRF9131-EK are representative of the nRF91 series.

A noteworthy point is that an nRF91 "chip" is in fact a System-in-Package (SiP): multiple dies wire-bonded together in a single package. This approach

allows for greater flexibility as different dies can be included or excluded, but the resulting products are more costly as more complex to manufacture.

We are conducting our experimental work with nRF91 chips. As is allowed in the standard, they only implement a subset of the PHY configurations of the DECT NR+ standard. That is, μ is always equal to 1, resulting in a subcarrier spacing of 27 kHz. Similarly, β is also always equal to 1, which means there always are 64 subcarriers, from which 56 are occupied and 8 acting as guard band. Finally, since nRF91 radios supports frequency bands 1, 2, 9, they can operate in all regions. The radios can output up to +19 dBm, which is within regulatory limits. The packet length on nRF91 radios is limited to 8 slots (16 subslots), and the MCS can be set between 0 and 4. Table 1 summarizes the resulting packet lengths. Note that not all configurations in this table are achievable on nRF91 radios, since the maximum payload size is limited to 700 B.

POWER CONSUMPTION

Power consumption is a key parameter when evaluating DECT NR+ for applications where nodes are battery-powered.

We use the Nordic Power Profiler Kit (PPK) II to measure the power consumption of DECT NR+. The PPK can power the nRF9161 DK and display that board's power consumption. It is accurate enough to evaluate the current draw during transmit (TX) and receive (RX) operations. The PPK measures the current draw of the entire nRF91 System-in-Package, including the current drawn by the Central Processing Unit (CPU) and the radio frequency (RF) modules in the SiP.

We developed a PHY test firmware (FW) that triggers TX and RX operations based on serial commands from a computer. This command allows the computer to select the TX power, MCS, packet length and frequency. On the computer, we developed a Python program to issue those serial commands, looping through all combinatorial possibilities of these parameters. This automated testing framework is based on the PPK Python Application Programming Interface (API) [9].

We measure the power consumption of the nRF91 while it is running the `mfw-nr+_nrf91x1_1.0.2` RF modem FW version. The supply voltage is 3.7 V. While the current draw varies with supply voltage, the energy consumption remains more or less constant. In the recorded current profile, TX operations are clearly visible as peaks with sharply rising and falling edges. The duration of these peaks depends on the subslot length specified; they precisely match the time durations defined in the DECT NR+ standard.

When the nRF9161 is actively transmitting, it draws between 70 mA (at -40 dBm TX power) and 220 mA (at +19 dBm TX power) of current. When the nRF9161 is actively receiving, it draws 45 mA. We present measurements for the nRF9161-DK, but verified that we measure similar values on the nRF9151-DK.

We plot the charge drawn during a single TX operation in Fig. 1. Although illustrated for MCS 0, the values remain unchanged across different MCS configurations. As expected, the charge drawn increases linearly with payload duration. Since the TX power is specified on a logarithmic scale, while charge consumed increases as the transmit powers increases, it is not a linear relationship. That is, as shown for a packet length of 8 slots in Fig. 1, going from -40 dBm to +7 dBm only increases charge consumed by 70 μC , going from +7 dBm to +19 dBm increases by 400 μC . Figure 2 shows the charge consumed *per bit*. The plot shows the benefit of using a higher MCS scheme: more bytes fit into the same number of subslots as we increase MCS. The largest efficiency boost is when switching from MCS 0 to MCS 1. The plots for MCS 2, 3 and 4 do not cover the entire range of payload sizes not to exceed the 700 B payload limit on the nRF91. Figure 2 also shows that it is more beneficial to use larger payload sizes rather than fragmenting. While in all cases the charge per bit decreases with increasing the number of bits, this is less true for higher MCS values (i.e. the lines in Fig. 2 “flatten” for higher MCS values). For instance, the charge drawn per bit in a packet transmitted at +19 dBm with MCS 0 reduces from 0.68 μC to 0.36 μC when going from a payload size of 2 subslots to 16 subslots. We can further see that the curves for MCS 4 are close, showing that, for a high MCS index, the impact of TX power on the efficiency per bit is small.

COMMUNICATION RANGE

Another important key performance indicator is communication range. Through a comprehensive measurement campaign, we measure the PDR as a function of distance between transmitter and receiver nodes.

For all range tests, we deploy nRF9131-EK boards using MCS 0, frequency channel number 1657 (1881.792 MHz) and a packet length of 5 subslots. The PDR is measured over 10,000 transmitted packets per TX power, over a duration of 4 hours

The firmware we developed for the nodes support two modes: initiator and responder. At the initiator, a computer sends commands to the nRF9131 over a serial port to execute transmit operations with different parameters. After the initiator transmits a wireless frame, it waits for an acknowledgment (ACK). The device in Initiator mode is marked as TX in the maps of Fig. 3. We call the other devices the “responder” nodes: they continuously listen for frames and acknowledge them. This setup allows us to record packet statistics at the initiator and several responder devices.

Figure 3 shows the four test environments.

Setup (a) is the Siemens Industrial Connectivity Lab (ICL) in Erlangen. The ICL is a factory hall designed for the evaluation of industrial IoT, and is characterized by a metal-rich environment that causes signal reflections and multipath fading. This makes it an ideal testbed for assessing wireless communication reliability in industrial settings.

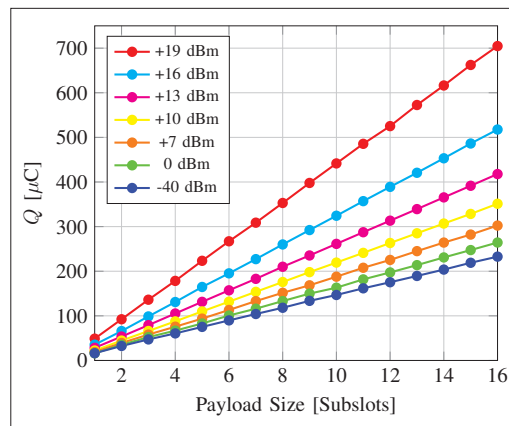


FIGURE 1. Average charge drawn during a TX operation, as measured on the nRF9161-DK.

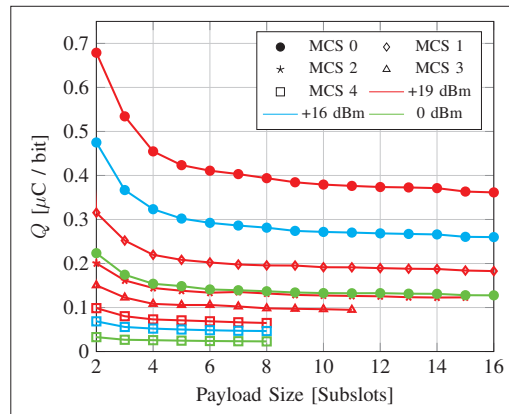


FIGURE 2. Average charge drawn per bit during a TX operation, as measured on the nRF9161-DK.

Setup (b) is an office environment, chosen to evaluate the penetration of walls and objects. This setup reflects real-world indoor deployments where wireless signals must navigate cubicles, furniture, and electronic devices, providing insights into indoor coverage, interference, and connectivity stability.

Setup (c) is the Siemens Campus in Erlangen, Germany, representing an urban outdoor environment. It allows us to assess the suitability of DECT NR+ in smart city applications, where high-rise buildings, moving vehicles, and dense infrastructure impact signal propagation and network reliability. Setup (d) is located on top of an old castle ruin with an extensive view of the surrounding rural area. The objective is to evaluate the performance of DECT NR+ in ideal LOS scenarios with minimal interference, making it representative of applications such as environmental monitoring, remote sensor networks, or long-range webcam image transmission.

Figure 4 shows the PDR results of the four test setups.

Setup (a) delivers solid performance despite limited line-of-sight (LOS) and no antenna alignment. Except for device a1, all nodes were in non-LOS, reflecting realistic deployment conditions. A transmit power of -8 dBm was sufficient to maintain a PDR above 80%. As expected, devices a2 and a4 show a gradual PDR decline followed by a steep drop-off. In contrast, a3 exhibits an unusual dip at 0 dBm, likely caused by multi-path interference from nearby metal surfaces [10].

Through a comprehensive measurement campaign, we measure the PDR as a function of distance between transmitter and receiver nodes.

Comparable detailed charge measurements have been conducted for the LTC5800 chip, which is part of Analog Devices' "SmartMesh" product line and based on IEEE 802.15.4 O-QPSK.

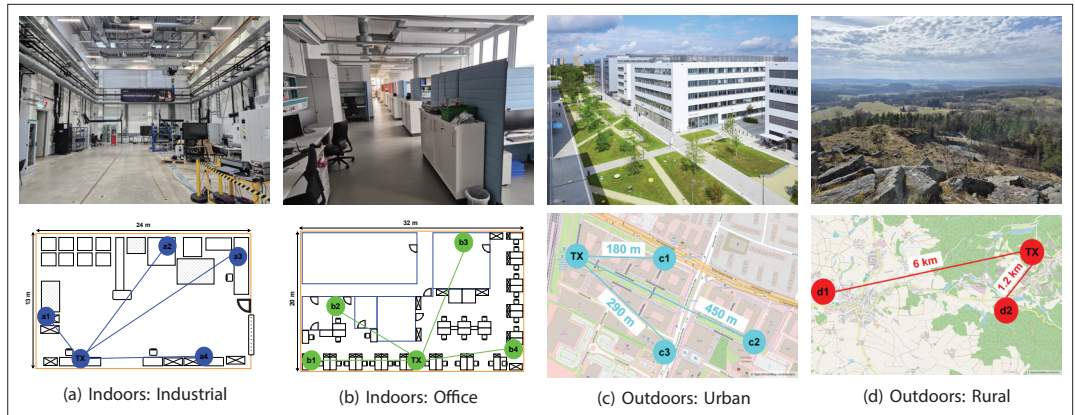


FIGURE 3. The four range test setups (top: communication environment, bottom: map with relative position and distances).

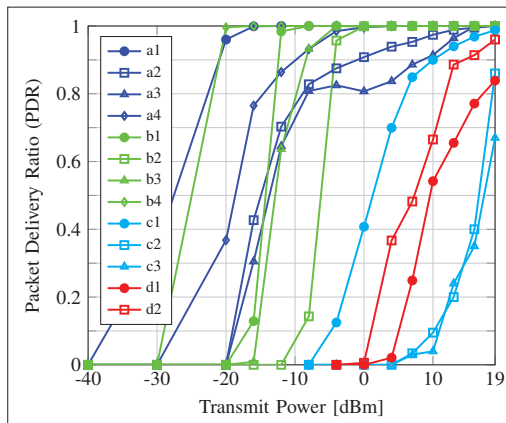


FIGURE 4. PDR evaluated for different transmit powers.

Setup (b), although entirely non-LOS, achieves strong PDR values near 100 % down to -4 dBm. However, unlike (a), the PDR drops sharply rather than gradually, likely due to the coarse granularity of the available low-power transmission levels and the restrictive indoor environment with thick walls and narrow corridors.

Setup (c) shows weak results despite elevated antenna placement and partial LOS for c2. While c1 performs reasonably well at 180 m through an urban canyon, c3 (290 m) struggles due to obstructions. Surprisingly, c2 underperforms despite a theoretically clear 450 m street path. Although its Receive Signal Strength Indicator (RSSI) values are better than c3's, its link quality is impaired by transient obstructions like vehicles and pedestrians.

Setup (d) demonstrates good long-range performance. Both d1 and d2 benefit from clear LOS to an elevated base station. While d2 maintains a strong connection at 1.2 km, d1 remains stable even at 6 km. These results exceed previously published DECT NR+ range measurements [11] and highlight how DECT NR+ can be used for remote and outdoor IoT scenarios where line-of-sight conditions are met.

So, Is DECT NR+ Any Good?

PUTTING OUR POWER CONSUMPTION RESULTS INTO PERSPECTIVE

The power consumption measurements mentioned earlier reveal current draws between 70 mA and 220 mA during transmission, and 45 mA during reception. These values are notably high when comparing to other technologies. As an

element of comparison, the nRF52840, a popular Bluetooth Low Energy (BLE) radio also from Nordic Semiconductor representative of the state of the art, draws 4.8 mA when transmitting at 0 dBm, 4.6 mA when receiving. Of course, DECT NR+ can be configured to transmit faster than BLE, in which case the duration the radio is on (and drawing current), is shorter.

At the maximum TX power of +19 dBm with a 16-subslot payload (249 B using MCS 0) a charge around 700 μ C is drawn in a single TX operation. This is a lot of charge, which hinders the wide adoption of DECT NR+ by applications with battery-powered devices.

Let's put that number into perspective. Comparable detailed charge measurements have been conducted for the LTC5800 chip, which is part of Analog Devices' "SmartMesh" product line and based on IEEE 802.15.4 O-QPSK. For sending a 127-byte frame at +8 dBm, this chip draws about 45 μ C. Transmitting a similar payload at +7 dBm with DECT NR+ using 9 subslots at MCS 0 draws 171 μ C.

It is possible to lower the power consumption per bit by tuning the MCS: if setting MCS to 4, only 2 subslots are needed, resulting in 39 μ C of charge needed. But this efficiency comes at a cost: our experience from further experiments tell us that increasing the MCS significantly decreases PDR and communication range. An IEEE 802.15.4-based solution likely has higher PDR for the same distance than DECT NR+ with MCS 4, potentially making it more energy efficient as fewer retransmissions are needed.

PUTTING OUR COMMUNICATION RANGE RESULTS INTO PERSPECTIVE

The range performance results from an earlier section are promising. We need to keep in mind, however, that these results are obtained with MCS 0, i.e. when using the maximum 16 subslot packet length (249 bytes) on the nRF, a data rate of 598 kbit/s is achievable.

In both indoor settings (industrial and office), we see that PDR above 80% is achieved with output power above -4 dBm. These good results can in part be attributed to the fact that DECT NR+ operates in the 1.9 GHz frequency band that is mostly interference free, unless of course DECT phones or other DECT NR+ networks operate nearby. The indoor communication coverage exceeds what we typically see with IEEE 802.15.4 O-QPSK or Bluetooth based solutions. DECT NR+ occasionally even maintains a stable connection,

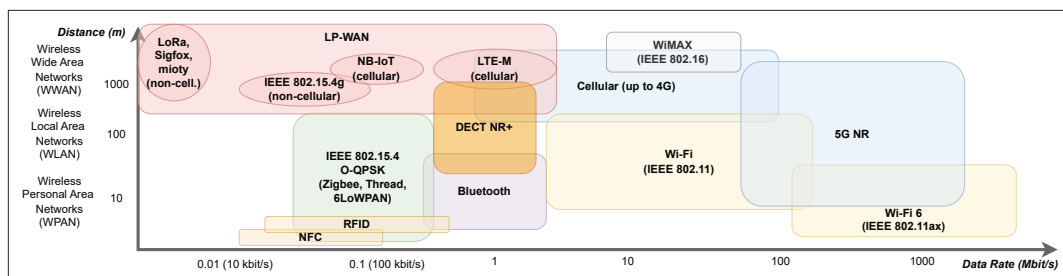


FIGURE 5. Classification of DECT NR+ among other standards.

whereas a smartphone connected point-to-point to a Wi-Fi router, operating at +20 dBm, had already lost connectivity under the same conditions.

In a perfectly line-of-sight scenario, tells us that DECT NR+ can achieve 6 km communication range. This is comparable to IEEE 802.15.4g [12], but does not reach the communication range of LoRa. Of course, even if LoRa is longer range, it has a much lower data rate and is subject to strict duty cycle regulations [13]. DECT NR+ therefore appears usable for long range applications, as long as line-of-sight conditions are met. For use cases where DECT NR+ satisfies the range scenarios, it has a clear advantage of LoRa as its data rate is higher, even at MCS 0.

Unfortunately, the range of DECT NR+ drops significantly in non-line-of-sight scenarios. As shown in our urban tests, non-LOS connections longer 200 m do not offer reliable communication. Technologies such as IEEE 802.15.4g and LoRa appear more appealing in urban environments as they offer better penetration as they operate in sub-GHz frequencies.

COMPARING DECT NR+ TO OTHER WIRELESS TECHNOLOGIES

Figure 5 is our attempt at placing DECT NR+ on a distance/data rate continuum, together with other wireless technologies. Creating such a plot is necessarily imperfect and cannot represent technical subtleties, but it is a fair way to get a big picture view of the complementarity of solutions. Importantly, it not depict strict performance boundaries, rather illustrates typical operating points. Ding *et al.* [14] already proposed a similar visualization. Our classification is supported by their survey and our own practical experience with the respective technologies. It is also a good visual foundation for the discussion that follows.

Can DECT NR+ be classified as a Wireless Personal Area Network (WPAN)? Figure 5 clearly shows that DECT NR+ lies close to Bluetooth/BLE and low-rate WPAN standards based on IEEE 802.15.4 O-QPSK, such as Zigbee or Thread. With the right settings, DECT NR+ can offer a data rate up to 3.36 Mbit/s, so faster than 2 Mbit/s BLE, and much faster than 250 kbit/s IEEE 802.15.4 O-QPSK. The difference in data rate compared to BLE is hence more incremental than disruptive. While the key differentiator between DECT NR+ and the mentioned WPAN standards is the larger achievable range, this advantage diminishes when using higher MCS settings required for increased data rates. Combined with significantly higher power consumption and cost, DECT NR+ is unlikely to replace WPAN standards, as its primary benefit of extended range becomes less compelling in high-throughput scenarios.

Can DECT NR+ be classified as a Wireless Local Area Network (WLAN)? IoT applications that require higher data rates than typical WPANs while maintaining a similar range often turn to Wi-Fi. This is largely due to Wi-Fi's use of significantly wider channel bandwidths. Given that Wi-Fi's data rates are substantially higher, particularly in recent revisions, classifying DECT NR+ as a WLAN technology would be misleading.

Can DECT NR+ be classified as a Low Power Wide Area Network (LPWAN)? With a 6 km communication range in the most favorable conditions, DECT NR+ does challenge traditional LPWAN standards. Although it offers far higher data rates as non-cellular LPWAN, DECT NR+ does not offer the extensive range of LoRa or Sigfox. And, although it offers a communication range comparable to IEEE 802.15.4g, it suffers from much higher power consumption and price. In terms of data rates, DECT NR+ aligns more closely with cellular LPWAN standards like LTE-M, while offering the distinct advantage of being non-cellular, thus avoiding cellular network subscriptions and their associated costs.

Can DECT NR+ keep up with 5G NR? The question might be odd, but it is important to answer clearly, since DECT NR+ is marketed as "the first non-cellular 5G standard". DECT NR+ is standardized by ETSI in the context of overarching 5G standardization activities, but we believe the statements about DECT NR+ and 5G are misleading. To most of us, the term "5G" refers to 5G NR, offering the high-bandwidth constant connectivity of our smartphones. We believe that marketing DECT NR+ as a "non-cellular 5G standard" conveys the notion that DECT NR+ somehow "goes beyond" 5G, maybe even suggesting DECT NR+ is a new version of 5G that gives our smartphones the same connectivity without needing to connect to the cellular base stations. This is, of course, not at all the case. DECT NR+ is not in any way in the technological realm of that definition of 5G as that of our smartphones: it is a reuse of legacy DECT cordless phone frequency band for IoT-type application.

So, where does DECT NR+ fit in? DECT NR+ can be positioned as a technology that fills a gap between cellular-LPWAN technologies, WPAN standards and Wi-Fi.

WHERE SHOULD I USE DECT NR+?

The lesson learned earlier is that DECT NR+ fills a narrow technological gap. And there are applications that require just that, while not being cost sensitive or needing a decade of battery-powered operation. One example is neighborhood-scale mesh networks to connect electricity meters,

While the key differentiator between DECT NR+ and the mentioned WPAN standards is the larger achievable range, this advantage diminishes when using higher MCS settings required for increased data rates.

which is exactly what company Wirepas successfully uses DECT NR+ for.

This is a rather narrow gap. We believe that what makes DECT NR+ really stand out is that it offers something none of the technologies above offers: *flexibility*. The physical layer of DECT NR+ allows devices to dynamically trade-off data rate for communication range. This can be interesting in at least the following use cases. First, a company might be designing an advanced mesh networking protocol in the spirit of g6TiSCH [15], that matches the PHY-layer settings dynamically on a link-by-link basis. Using radios such as the nRF91 series is ideal in that case, as the networking stack can dynamically set the modulation settings detailed earlier. Second, and maybe more importantly, a company might be designing wireless networks for a range of applications. Some of these applications might require long range, others high throughput. Using DECT NR+ then comes in very handy as different types of applications can be addressed without requiring any redesign.

THE ROAD AHEAD FOR DECT NR+

Nordic Semiconductor already offers commercially available DECT NR+-compatible chips. This availability facilitates both research and broader adoption of the technology. Companies like Wirepas, Stratum 9, Last Mile Semiconductor, and Sennheiser are investing in the technology. While the ecosystem is still very young, we expect it to grow. We expect other chip vendors to add DECT NR+-compliant chips to their catalog. Competition will drive prices and power consumption down, although, given the complex nature of the PHY, it is unlikely DECT NR+ chips will be competitive on those fronts with traditional WPAN or LPWAN technologies. Whether DECT NR+ will ever be low-power enough to enable battery-powered applications is an open question. On the software front, we expect mesh networking protocols to be developed on top of DECT NR+, driven by open-source communities and commercial entities. Having such implementations will allow quicker evaluation of the technology. On the research front, we expect studies to be published about the applicability of DECT NR+ to specific use cases, in particular Industrial IoT, as well as studies on the coexistence between DECT NR+ and DECT phones.

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REFERENCES

- [1] D. Wang et al., "From IoT to 5G HoT: The Next Generation IoT-Based Intelligent Algorithms and 5G Technologies," *IEEE Commun. Mag.*, vol. 56, no. 10, 2018, pp. 114–20.
- [2] M. Penner, M. Nabeel, and J. Peissig, "URLLC Performance Evaluation of IMT-2020 Candidate Technology: DECT-2020 New Radio," *IEEE Vehic. Tech. Conf. (VTC)*, 2021.
- [3] R. Kovalchukov et al., "DECT-2020 New Radio: The Next Step toward 5G Massive Machine-Type Communications," *IEEE Commun. Mag.*, vol. 60, no. 6, 2022, pp. 58–64.
- [4] J. Dürre and N. Werner, "DECT-2020 NR for Professional Live Audio: Design Space Exploration and Practical Evaluation," *IEEE Int'l. Symp. Internet of Sounds (IS2)*, 2024.
- [5] E. Llaguno et al., "Physical Layer Performance of DECT-2020 New Radio for Factory Automation," *IEEE Int'l. Conf. Factory Comm. Systems (WFCS)*, 2024.
- [6] T. Nihtilä and H. Berg, "Energy Consumption of DECT-2020 NR Mesh Networks," *Joint European Conf. Networks and Communications and 6G Summit (EuCNC/6G Summit)*, 2022.
- [7] DECT-2020 New Radio (NR) Part 1-5 Release 2, European Telecommunications Standards Institute (ETSI) Std. ETSI TS 103 636-1/2/3/4/5 V2.1.1, Oct. 2024.
- [8] M. Waßmann et al., "Performance of DECT-2020 NR in An Industrial Environment for Varying RF Bands," *Joint European Conf. Networks and Communications and 6G Summit (EuCNC/6G Summit)*, 2024.
- [9] Institute for Development of Advanced Applied Systems (IRNAS), "PPK II Python API," 2023; <https://github.com/IRNAS/ppk2-api-python>.
- [10] V. C. Gungor and G. P. Hancke, "Industrial Wireless Sensor Networks: Challenges, Design Principles, and Technical Approaches," *IEEE Trans. Industrial Electronics*, vol. 56, no. 10, 2009, pp. 4258–65.
- [11] M. M. Haque et al., "DECT-2020 NR Link Distance Performance in Varying Environments: Models and Measurements," 2024; <https://arxiv.org/abs/2410.24112>.
- [12] J. Muñoz et al., "Evaluation of IEEE802.15.4g for Environmental Observations," *MDPI Sensors*, vol. 18, no. 10, 2018.
- [13] F. Adelantado et al., "Understanding the Limits of LoRaWAN," *IEEE Commun. Mag.*, vol. 55, no. 9, 2017, pp. 34–40.
- [14] J. Ding et al., "IoT Connectivity Technologies and Applications: A Survey," *IEEE Access*, vol. 8, 2020, pp. 67,646–73.
- [15] M. Rady et al., "g6TiSCH: Generalized 6TiSCH for Agile Multi-PHY Wireless Networking," *IEEE Access*, vol. 9, 2021.

BIOGRAPHIES

FABIAN GRAF (fabian.graf@siemens.com) is a Ph.D. candidate at Sorbonne University in Paris, France and works at Siemens AG in Erlangen, Germany. He holds a M.Sc. in Electrical Engineering from TU Munich, Germany. His research focuses on low-power wireless communication systems.

THOMAS WATTEYNE (thomas.watteyne@inria.fr) is a Senior Research Director at Inria Paris, France. He is currently on a sabbatical and works as a Principal Engineer at Analog Devices in Boston, USA. He holds a Ph.D. degree in Computer Science from INSA Lyon, France. He was a postdoctoral research lead in Prof. Kristofer Pister's team with the University of California at Berkeley, USA. From 2005 to 2008, he was a Research Engineer with France Telecom, Orange Labs. From 2023 to 2024, he served as the Scientific Coordinator of the Horizon Europe OpenSwarm Project. From 2013 to 2022, he co-chaired the IETF 6TiSCH Working Group to standardize how to use IEEE802.15.4e TSCH in IPv6-enabled mesh networks.

MICHAEL VILLNOW (michael.villnow@siemens.com) leads the research group for sensor systems at Siemens AG in Erlangen, Germany. He received the Diploma degree in Mechatronics from Friedrich-Alexander-University (FAU) of Erlangen-Nuremberg, Germany. Since 2015, he has been a Project Manager at Siemens for IoT development projects.